Answer **all** the questions in this Section in the spaces provided.

1 A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 1.1.

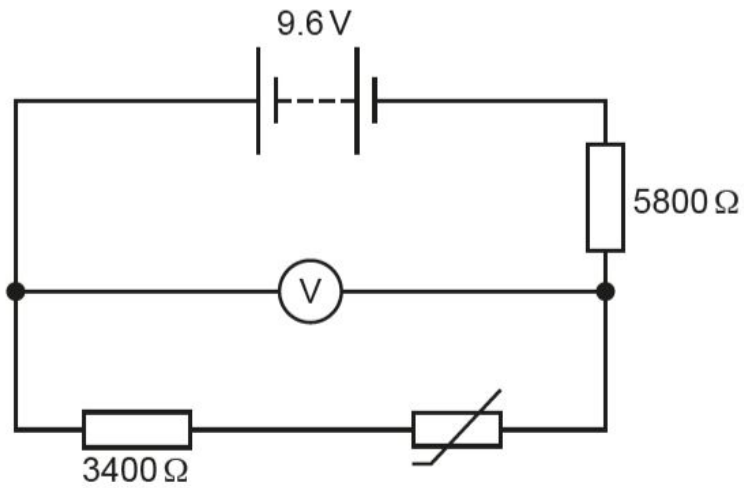


Fig. 1.1

The fixed resistors have resistances of 3400 Ω and 5800 Ω. The reading on the voltmeter in the circuit is 6.0 V.

(a) Calculate the current in the resistor of resistance 5800 Ω.

$9.6 = 6.0 + (I \times 5800)$ or $3.6 = I \times 5800$ [C1]

$I = 6.2 \times 10^{-4}$ A [A1]

current = ... A [2]

(b) Calculate the resistance of the thermistor.

$9.6 = 6.2 \times 10^{-4} \times (3400 + 5800 + R)$ [C1]

or

$6.0 = 6.2 \times 10^{-4} \times (3400 + R)$

$R = 6.3 \times 10^{3}\ \Omega$ [A1]

resistance = ... Ω [2]

(c) The initial energy stored in the battery is 2.6×10^4 J.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

$(\Delta W = \varepsilon \times Q =) 9.6 \times 330 (= 3170 \text{ J})$ [C1]

final stored energy = $2.6 \times 10^4 - 3170$
$= 2.3 \times 10^4$ J [A1]

final stored energy = ... J [2]

(d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

(i) the temperature of the thermistor

decrease [B1]
(since an NTC thermistor increases in resistance when temperature falls) . [1]

(ii) the current in the thermistor

decrease [B1]
. (since total resistance increases with the e.m.f. fixed) ... [1]

(iii) the potential difference across the thermistor.

increase [B1]
... (p.d. across the fixed resistors decreases, and since the total e.m.f. is fixed, the p.d. across the thermistor increases) ... [1]

(e) If the temperature of the thermistor in Fig. 1.1 changes at a constant rate, state a reason why the potential difference across the thermistor does not change at a constant rate.

.. change of resistance with temperature is not linear (Thermistor is a non-ohmic conductor) [B1] .....
OR
.. potential divider equation is non-linear .....

..... [1]

[Total: 10]